Maheshkumar Madipeddi

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PROFESSIONAL SUMMARY

Data Analyst with 4+ years of experience specializing in statistical modeling and data mining to drive business intelligence. Proficient in executing complex data workflows using Python, SQL, and R, with a deep focus on data cleansing and transformation across diverse datasets. Expertise includes applying causal inference and predictive modeling to identify performance trends and automate reporting processes via Apache Airflow and Tableau. Skilled in utilizing AWS and Google Cloud Platform to manage large-scale database systems while delivering actionable insights through advanced visualization tools like Power BI and Matplotlib. Committed to enhancing operational efficiency by developing automated analysis templates and maintaining high-integrity data pipelines that support informed decision-making for technical and non-technical leadership.

TECHNICAL SKILLS

Data Engineering: Apache Airflow, Talend, ETL Processes, Data Pipelines, Data Cleansing, Data Transformation

Cloud & Databases: AWS Lambda, AWS S3, Google Cloud Platform, Vertex AI, Redis, PostgreSQL, DBMS

BI & Visualization: Power BI, Tableau, Visio, Matplotlib, Seaborn

Programming & Languages: Python (Pandas), SQL, R, Java, PySpark

Productivity & Design Tools: Microsoft Excel, Microsoft PowerPoint, Microsoft Word, Git, Jupyter Notebooks

Statistical & Analytical Methods: Statistical Modeling, Causal Inference, A/B Testing, Data Mining, Bayesian Modeling, Predictive Modeling

Architecture & Advanced Analytics: RESTful APIs, Microservices, Data Warehousing, Automation Scripts, Trend Analysis, Regression Analysis

WORK EXPERIENCE

Aescape, Data Analyst□**Mar 2024 to Present**

- Developed foundational data tables using advanced SQL to standardize key metrics for wellness technology performance, resulting in better consistency across organizational reporting and decision-making processes.
- Built automated dashboards in Tableau that provided self-service analytics for operational teams to monitor daily equipment usage and system health without manual data pulls.
- Executed A/B tests for robotic massage features and analyzed the results using statistical methods in Python to guide product development and feature prioritization.
- Defined and tracked critical business KPIs using SQL to identify performance trends and surface anomalies in user engagement patterns within the wellness platform.
- Created customer satisfaction measurement initiatives to gather user feedback and reported key insights to the engineering team to improve hardware and software interactions.
- Improved data pipelines using Apache Airflow to handle larger datasets more efficiently as the user base for the AI-driven wellness technology expanded.
- Resolved complex business questions by performing detailed performance analysis with advanced SQL queries to identify opportunities for operational improvements.
- Maintained robust data infrastructure by monitoring pipeline health and integrating external vendor data using APIs to ensure a comprehensive view of business operations.
- Built automated analytics solutions using Python and Pandas which removed the need for manual data processing when generating monthly departmental performance reports.
- Improved data accuracy by diagnosing warehouse issues promptly and developing automated validation checks to ensure high-quality data for strategic planning.

LNT Machining Inc, Business Analyst□**Jun 2021 to Jul 2023**

- Designed complex SQL queries to extract and aggregate CNC machine performance data, creating foundational datasets that improved the accuracy of production efficiency reporting for manufacturing leadership.
- Automated the cleansing and transformation of raw sensor data from industrial equipment using Python scripts, ensuring high data quality for downstream statistical modeling and operational analysis.
- Implemented interactive Power BI dashboards that visualized real-time production metrics and scrap rates, allowing department heads to identify and address bottlenecks in the machining workflow.
- Applied statistical modeling techniques in R to analyze tool wear patterns and predict maintenance needs, reducing unplanned downtime and optimizing the manufacturing equipment lifecycle.
- Configured advanced Excel models using pivot tables and VLOOKUPs to perform ad-hoc analysis on supply chain costs, supporting data-driven decisions regarding raw material procurement.
- Managed data-focused projects by gathering business requirements from stakeholders and translating them into technical specifications for new database implementations and reporting enhancements.
- Built Talend workflows to integrate disparate data sources from inventory management and shop floor systems, providing a unified view of operational health and resource utilization.
- Created detailed process maps in Visio for CNC operations to identify manual gaps, which facilitated the design of automated data collection pipelines for production metrics.
- Developed trend analysis models using Python to evaluate the impact of process changes on part quality, applying causal inference to validate improvements in manufacturing precision.
- Enhanced data-driven decision-making by presenting strategic recommendations to leadership during weekly reviews, focusing on optimizing production output and reducing operational overhead through better visibility.

EDUCATION

Master of Science in Information Technology Management □July 2025

Lindsey Wilson University Columbia, KY